

CSA0302-DATA STRUCTURES

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C01-ASSESSMENT TOO1-3

Q1 Reverse a list

1 mark

Write a C program that takes an array as input and returns it reversed.

Your Code

```
1 #include <stdio.h>
2 int main() {
3     int n,i,a[100];
4     scanf("%d",&n);
5     for(i=0;i<n;i++)
6         scanf("%d",&a[i]);
7     for(i=n-1;i>=0;i--)
8         printf("%d",a[i]);
9     return 0;
10 }
```

Input

```
5
1 2 3 4 5
```

Output

```
54321
```

✔ Submitted

Q3 Searching

1 mark

Write a C program for Searching an Element in a Sorted Array

Your Code

```
1 #include <stdio.h>
2 int main() {
3     int a[]={10,20,30,40,50},
4     key = 30;
5     int l=0,h=4,m;
6     while (l<=h) {
7         m = (l + h) / 2;
8         if (a[m] == key) {
9             printf("found");
10            return 0;
11        }
12        if (a[m]<key) l = m+1;
13        else h=m-1;
14    }
15    printf("not found");
16    return 0;
17 }
```

Input

stdin input

Output

found

Q4 Searching

1 mark

Write a C program for Count the Occurrences of an Element

Your Code

```
1 #include <stdio.h>
2 int main() {
3     int a[]={1,2,3,2,2},key=2,i,c=0;
4     for(i=0;i<5;i++)
5         if(a[i]==key) c++;
6     printf("%d",c);
7     return 0;
8 }
```

Input

3

Output

3

 Submitted

Write a C program for Find Missing Element in an Array

Your Code

```
1 #include <stdio.h>
2
3 int main() {
4     int n, i, sum = 0, total = 0, a[100];
5
6     printf("Enter n: ");
7     scanf("%d", &n);
8
9     printf("Enter %d elements:\n", n - 1);
10    for(i = 0; i < n - 1; i++) {
11        scanf("%d", &a[i]);
12        sum += a[i];
13    }
14
15    total = n * (n + 1) / 2;
16
17    printf("Missing Element = %d", total - sum);
18
19    return 0;
20 }
```

Input

5
1 2 3 5

Output

Enter n: Enter 4 elements:
Missing Element = 4

Your Code

```
1 #include <stdio.h>
2
3 int main() {
4     int a[100], n, i, key;
5     int first = -1, last = -1;
6
7     printf("Enter number of elements: ");
8     scanf("%d", &n);
9
10    printf("Enter elements:\n");
11    for(i = 0; i < n; i++)
12        scanf("%d", &a[i]);
13
14    printf("Enter element to search: ");
15    scanf("%d", &key);
16
17    for(i = 0; i < n; i++) {
18        if(a[i] == key) {
19            if(first == -1)
20                first = i;
21            last = i;
22        }
23    }
24
25    printf("First Occurrence = %d\n", first);
26    printf("Last Occurrence = %d", last);
27 }
```

Input

6
10 20 30 20 40 20
20

Output

Enter number of elements: Enter
elements:
Enter element to search: First
Occurrence = 1
Last Occurrence = 5

Q8 Second Largest Element

1 mark

Write a C program for Find the Second Largest Element

Your Code

```
1 #include <stdio.h>
2
3 int main() {
4     int a[100], n, i, largest, second;
5
6     printf("Enter number of elements: ");
7     scanf("%d", &n);
8
9     printf("Enter elements:\n");
10    for(i = 0; i < n; i++)
11        scanf("%d", &a[i]);
12
13    largest = second = a[0];
14
15    for(i = 1; i < n; i++) {
16        if(a[i] > largest) {
17            second = largest;
18            largest = a[i];
19        } else if(a[i] > second && a[i] != largest) {
20            second = a[i];
21        }
22    }
```

Input

```
5
10 20 30 40 25
```

Output

```
Enter number of elements: Enter
elements:
Second Largest = 30
```

Q7 Largest and Smallest element

1 mark

Write a C program for Find the Largest and Smallest Element in an Array

Your Code

```
1 #include <stdio.h>
2
3 int main() {
4     int a[100], n, i, largest, smallest;
5
6     printf("Enter number of elements: ");
7     scanf("%d", &n);
8
9     printf("Enter elements:\n");
10    for(i = 0; i < n; i++)
11        scanf("%d", &a[i]);
12
13    largest = smallest = a[0];
14
15    for(i = 1; i < n; i++) {
16        if(a[i] > largest)
17            largest = a[i];
18        if(a[i] < smallest)
19            smallest = a[i];
20    }
```

Input

```
5
10 25 8 40 15
```

Output

```
Compilation Error:
C:\Windows\TEMP\sandbox_192511250
RC_6a6859d7a47924.64960841\main.c
: In function 'main':
C:\Windows\TEMP\sandbox_192511250
RC_6a6859d7a47924.64960841\main.c
```

Q6 Merging

1 mar

Write a C program for Merge Two Arrays

Your Code

```
1 #include <stdio.h>
2
3 int main() {
4     int arr1[100], arr2[100], merged[200];
5     int n1, n2, i;
6
7     // Input first array
8     printf("Enter the number of elements in first array: ");
9     scanf("%d", &n1);
10
11     printf("Enter the elements of first array:\n");
12     for (i = 0; i < n1; i++) {
13         scanf("%d", &arr1[i]);
14     }
15
16     // Input second array
17     printf("Enter the number of elements in second array: ");
18     scanf("%d", &n2);
19
20     printf("Enter the elements of second array:\n");
21     for (i = 0; i < n2; i++) {
```

Input

```
Enter the number of elements in
first array: 3
Enter the elements of first array:
10 20 30
```

Output

```
Enter the number of elements in
first array: Enter the elements
of first array:
Enter the number of elements in
second array: Enter the elements
of second array:
```

Q5 Array

1 mark

Write a program for binary search

Your Code

```
1 #include <stdio.h>
2
3 int main() {
4     int a[100], n, key, low, high, mid, i;
5
6     printf("Enter number of elements: ");
7     scanf("%d", &n);
8
9     printf("Enter sorted elements:\n");
10    for(i = 0; i < n; i++)
11        scanf("%d", &a[i]);
12
13    printf("Enter element to search: ");
14    scanf("%d", &key);
15
16    low = 0;
17    high = n - 1;
18
19    while(low <= high) {
20        mid = (low + high) / 2;
```

Input

```
Enter number of elements: 5
Enter sorted elements:
10 20 30 40 50
Enter element to search: 30
```

Output

Q2 Reverse a string

1 mark

Write a C program for palindrom

Your Code

```
1 #include <stdio.h>
2 int main() {
3     int n, r=0,t,d;
4     scanf("%d",&n);
5     t=n;
6     while (n) {
7         d=n%10;
8         r=r*10+d;
9         n /=10;
10    }
11    printf(t==r ? "palindrome" : "not palindrome");
12    return 0;
13 }
```

 Submitted

Input

121

Output

palindrome